Terms and Trends

Sparsh Mittal

Contents and objective

- Getting familiar with terms & concepts
- Overall trends
- Figure of merit (metrics)
- Computer design principles

Apollo 11 vs iPhone 10

- Apollo 11
  - Cost: ~$200 billion

  (whole program, adjusted)
- **Guidance Computer (1966):**
  - 2.048 MHz clock
  - 33,600 transistors
  - 85,000 instructions per second

- iPhone 10
  - Cost: ~$999

- **Apple A11 Bionic CPU (2017):**
  - 2390 MHz clock (~1000X faster)
  - 4.3 billion transistors (125,000X more)
  - 57 billion instructions per second (671,000X faster)
    - + Performs 600 billion neural network operations per second
    - + Processes 12 million pixels per second from still image camera
    - + Processes 2 million image tiles per second from record video (feature extraction, classification, and motion analysis)
    - + Encodes 240 frames per second at 1920x1080 (500 million pixels per second)

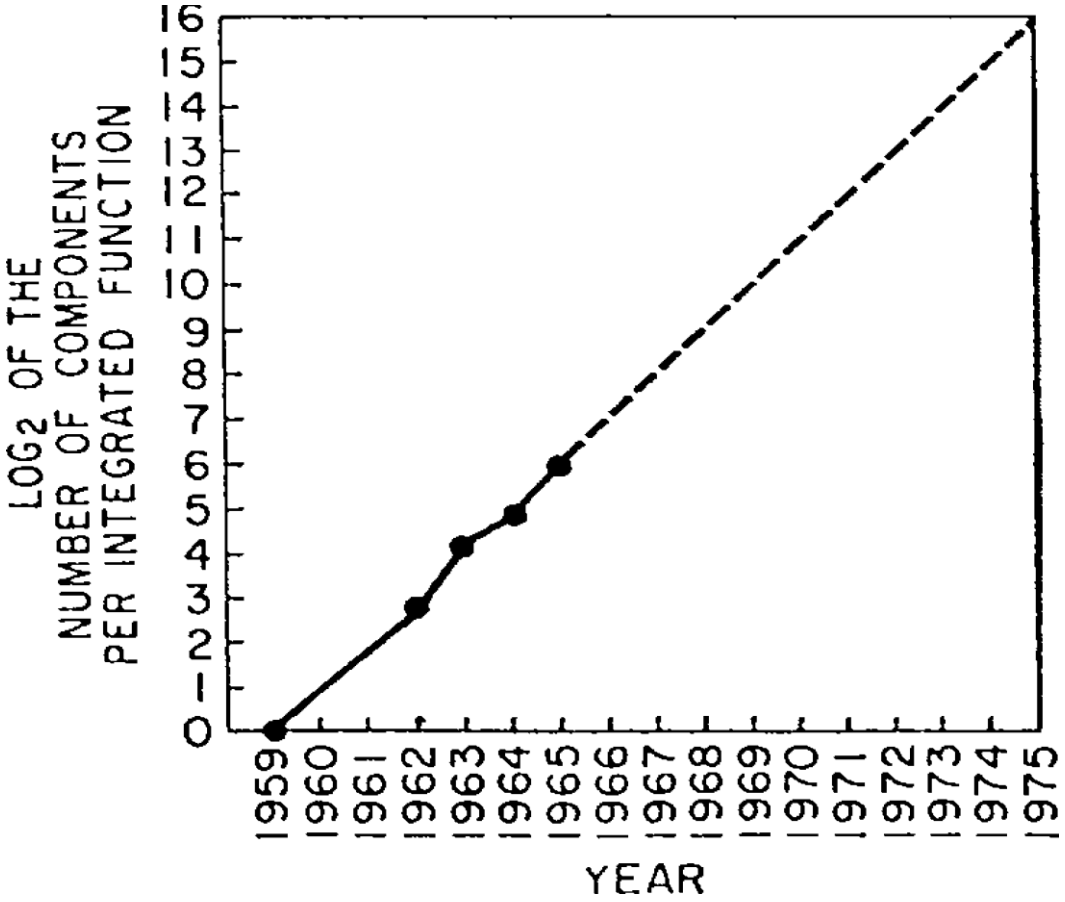
Courtesy: Jason Bakos

Table 1. Illustrative comparison of hardware for CloudML, MobileML and TinyML, including the MCUs targeted in this work.

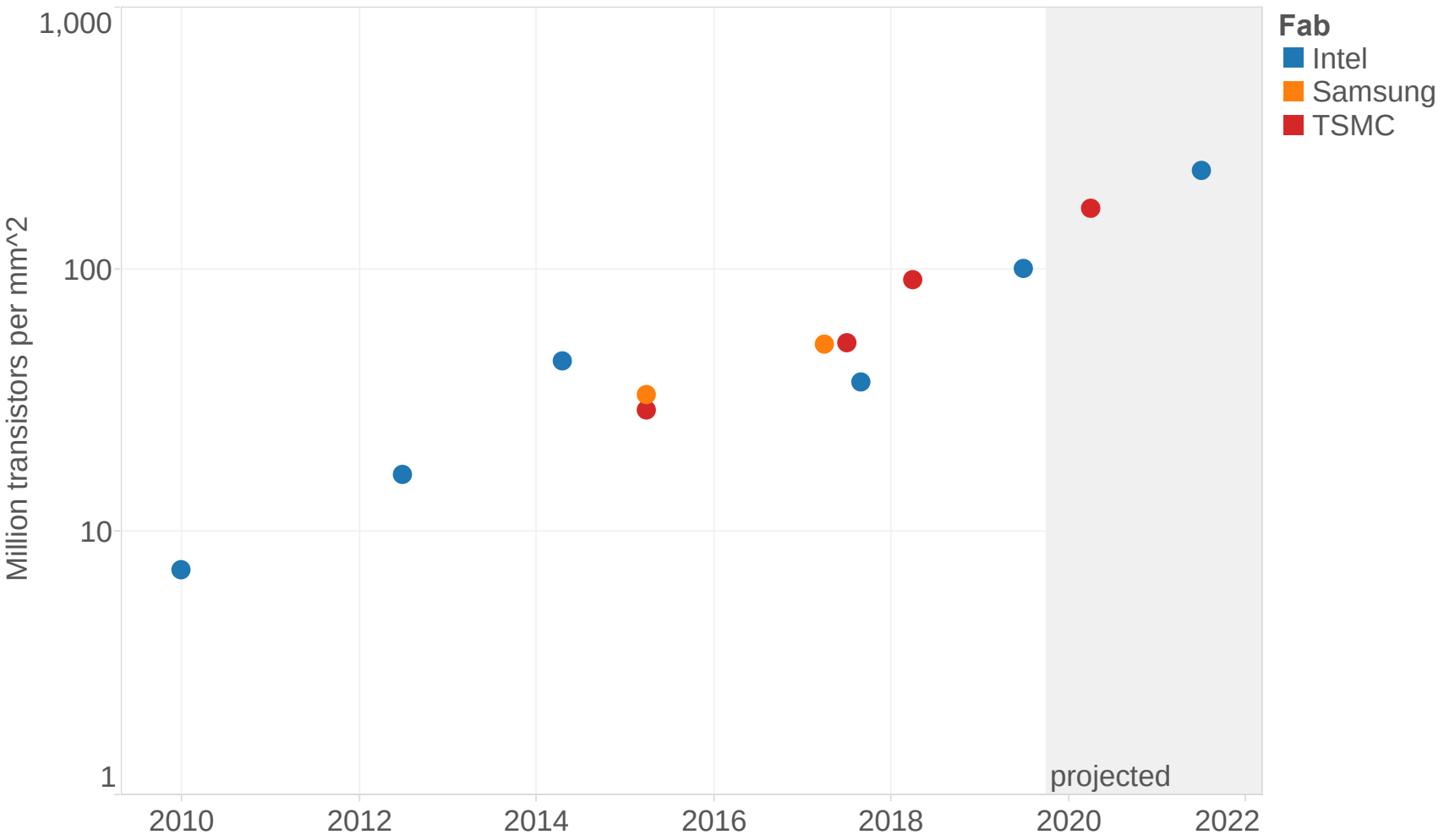
Platform	Architecture	Memory	Storage	Power	Price
CloudML	GPU	HBM	SSD/Disk		
Nvidia V100	Nvidia Volta	16GB	TB∼PB	250W	$9K
MobileML	CPU	DRAM	Flash		
Cell Phone	Mobile CPU	4GB	64GB	∼8W	∼$750
TinyML	MCU	SRAM	eFlash		
F446RE	Arm M4	128KB	0.5MB	0.1W	$3
F746ZG	Arm M7	320KB	1MB	0.3W	$5
F767ZI	Arm M7	512KB	2MB	0.3W	$8

Moore's Law - 1965

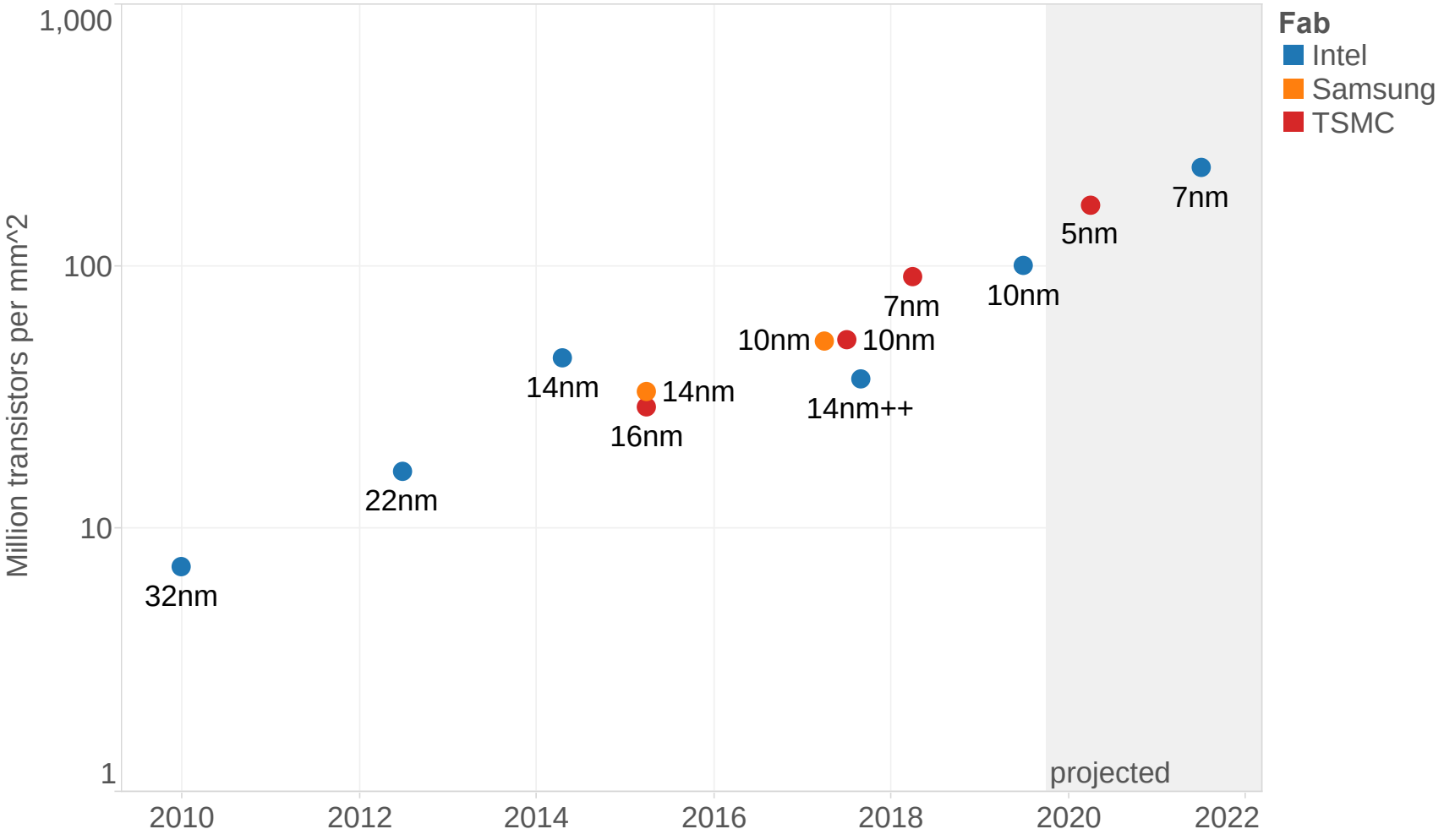
Moore's Law: transistors on a chip double every 18-24 months



Moore's Law today



Moore's Law today

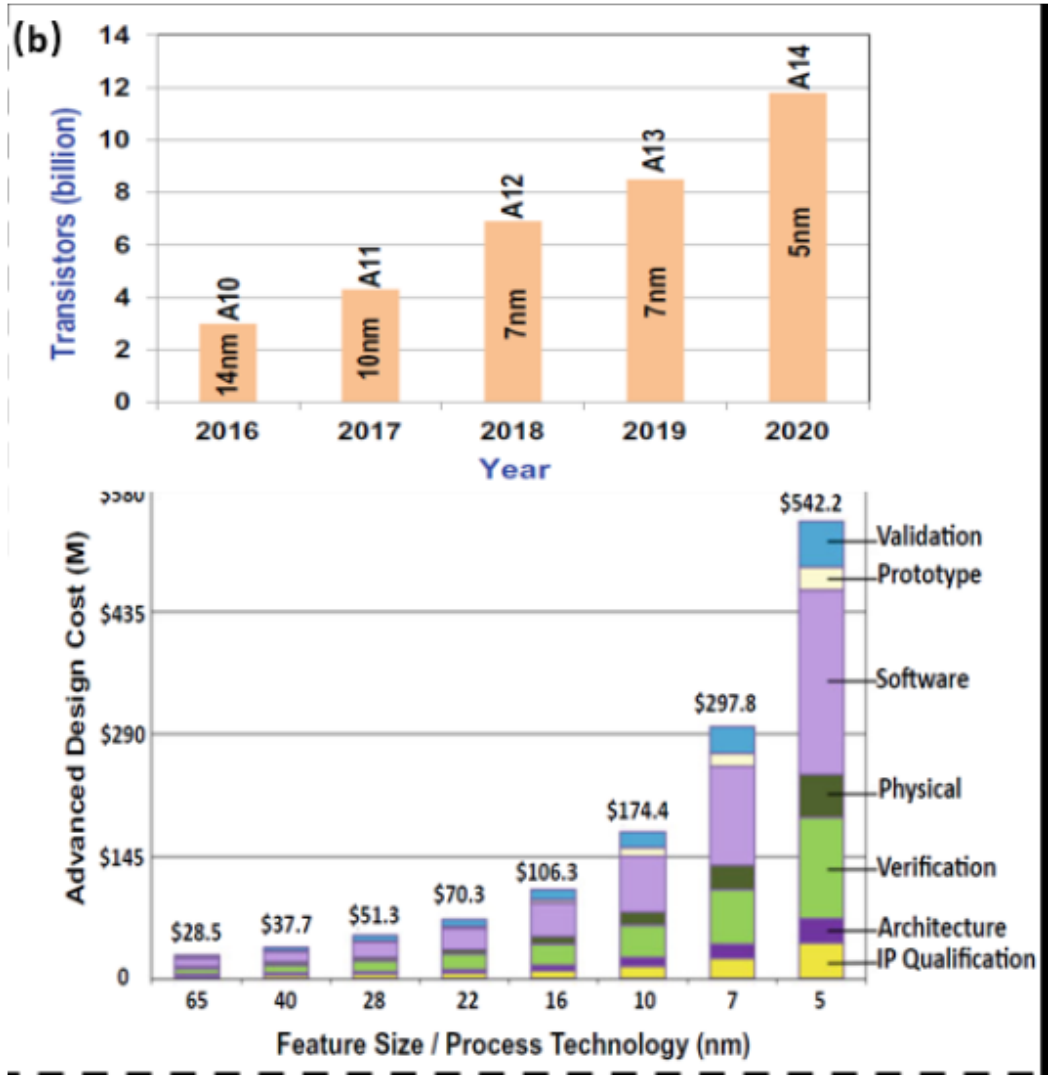


Transistor dimension (Area)

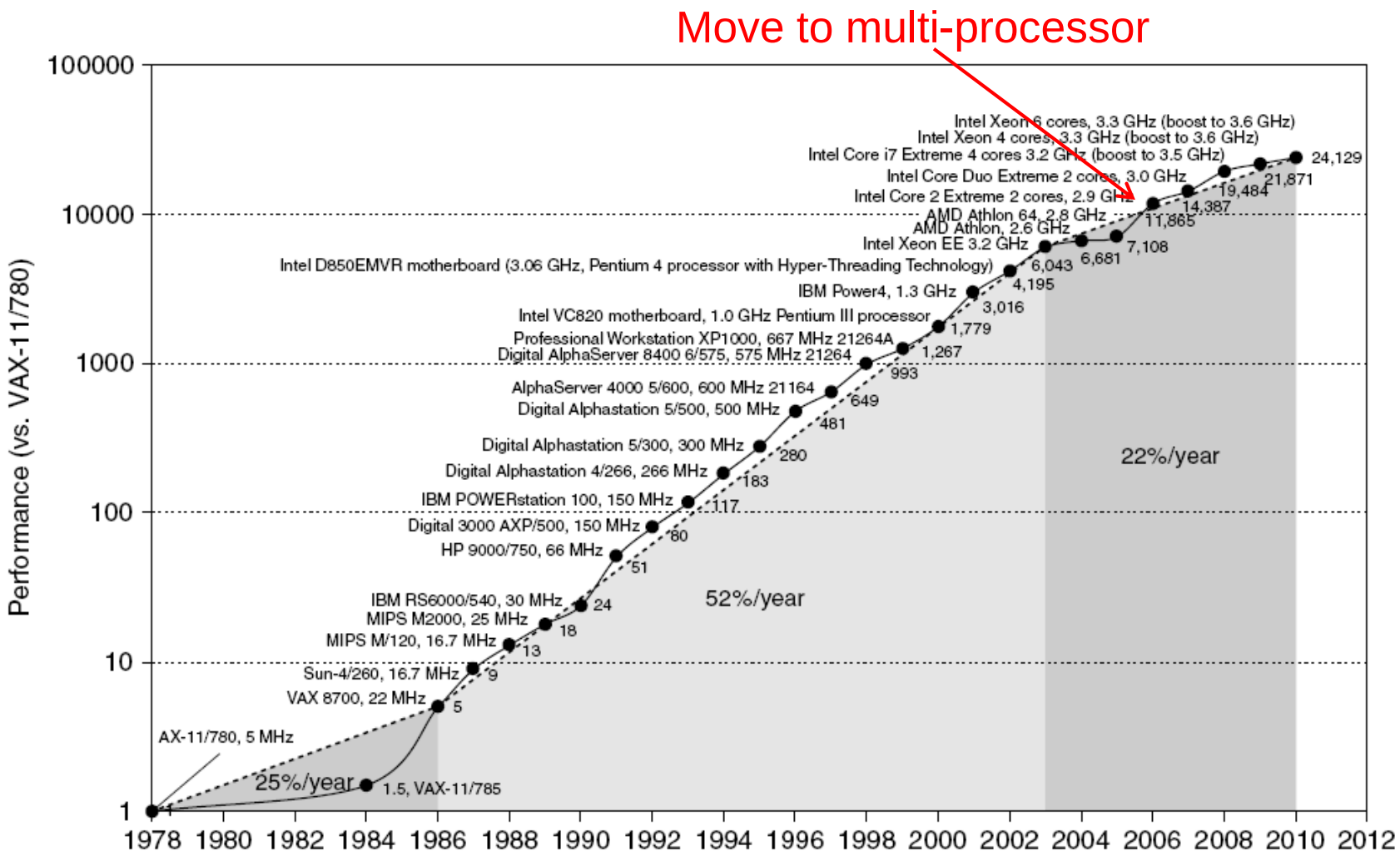
- Feature size
  - Also called geometry, process node
  - Minimum size of transistor or wire in x or y dimension

  - 10 microns in 1971 to .032 microns in 2011, .016 microns in 2016
  - Leads to chip-miniaturization
  - Allows fitting more transistors on a chip

iPhone #transistor and cost



Micro Processor Performance



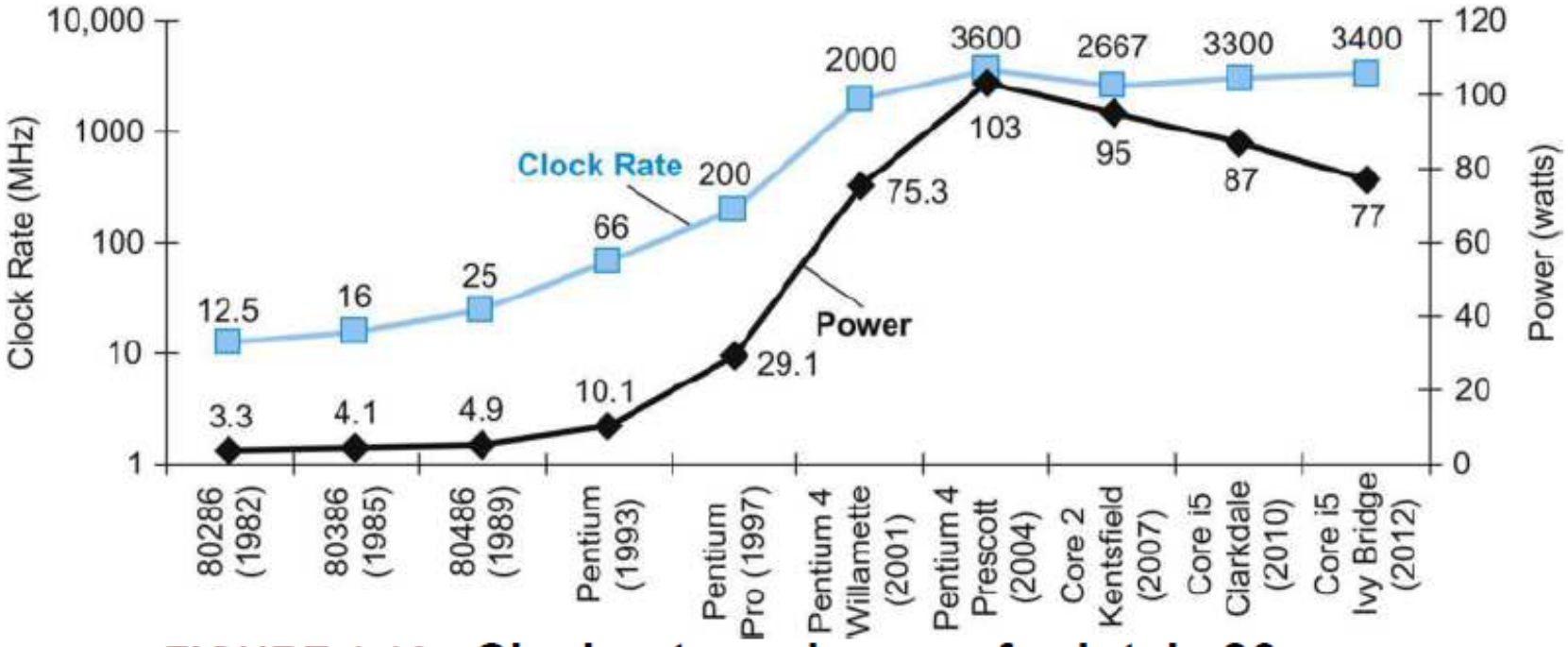
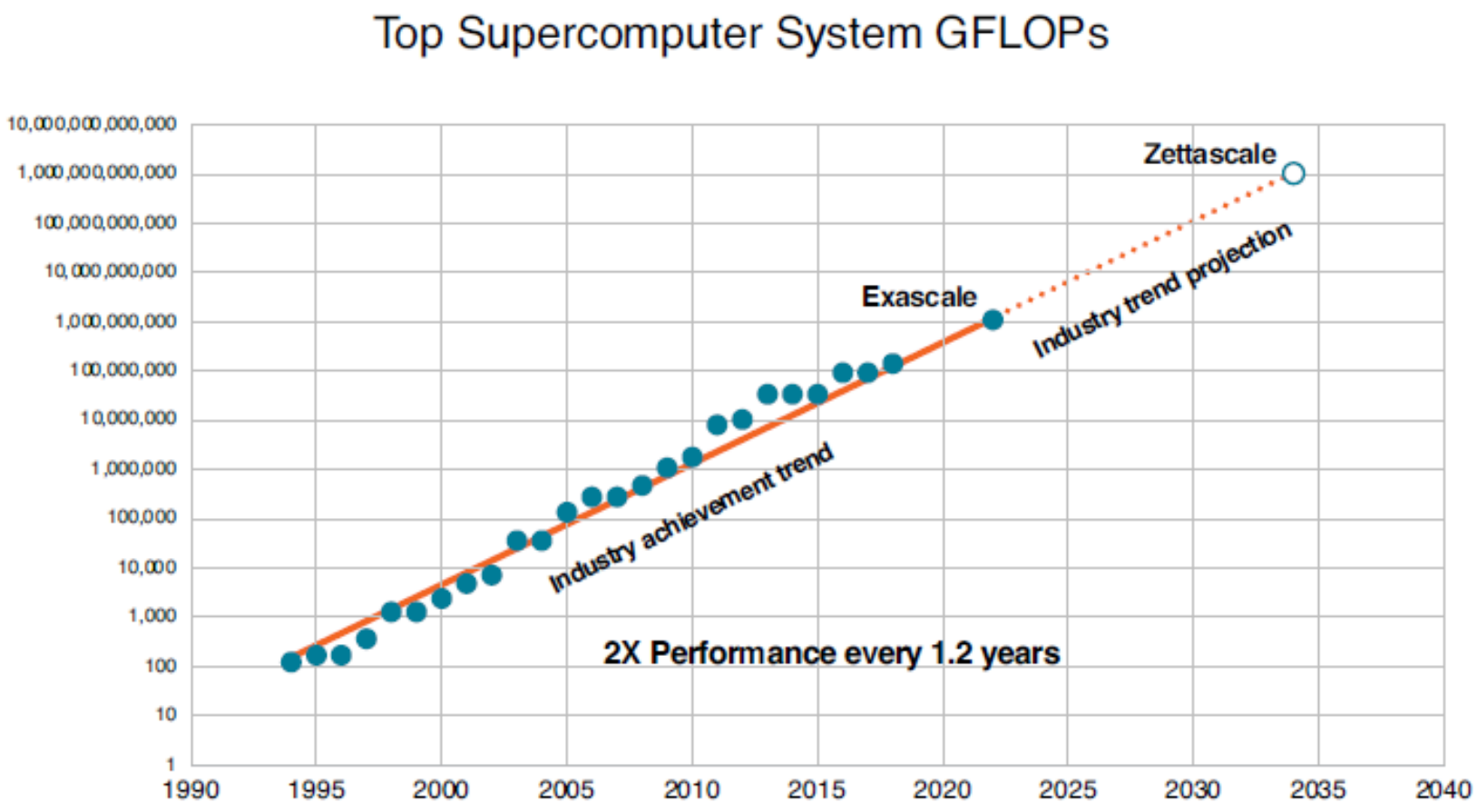


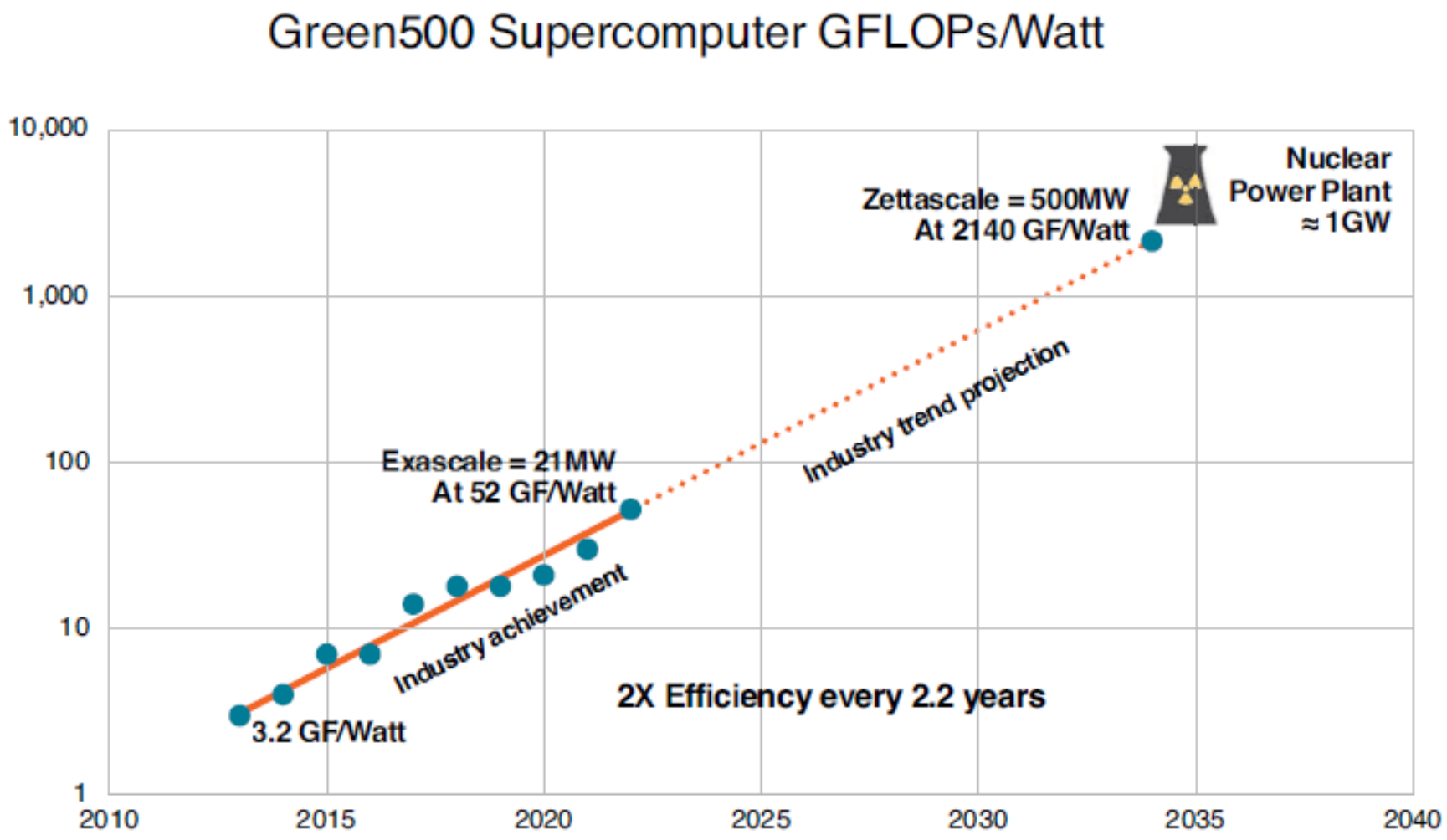
FIGURE 1.16 Clock rate and power for Intel x86 microprocessors over eight generations and 30 years.

The Pentium 4 made a dramatic jump in clock rate and power but less so in performance. The Prescott thermal problems led to the abandonment of the Pentium 4 line. The Core 2 line reverts to a simpler pipeline with lower clock rates and multiple processors per chip. The Core i5 pipelines follow in its footsteps.

Trend in High Performance Linpack Flops of top performing super-computers on top500.org list



Trend in power efficiency of top performing super-computers on top500.org list



Points to Note

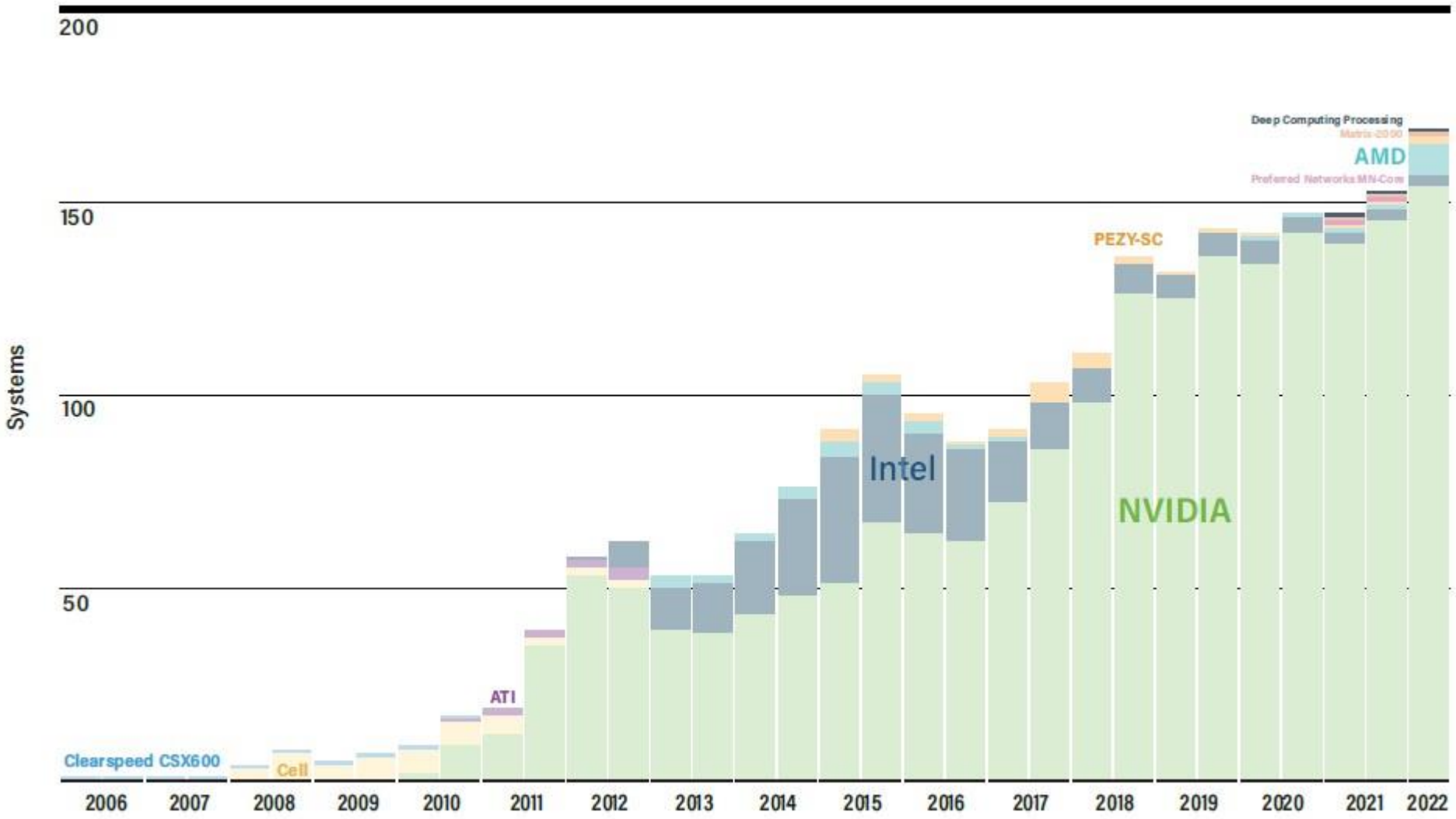
- The 52% growth per year is because of faster clock speeds and architectural innovations (led to 25x higher speed)

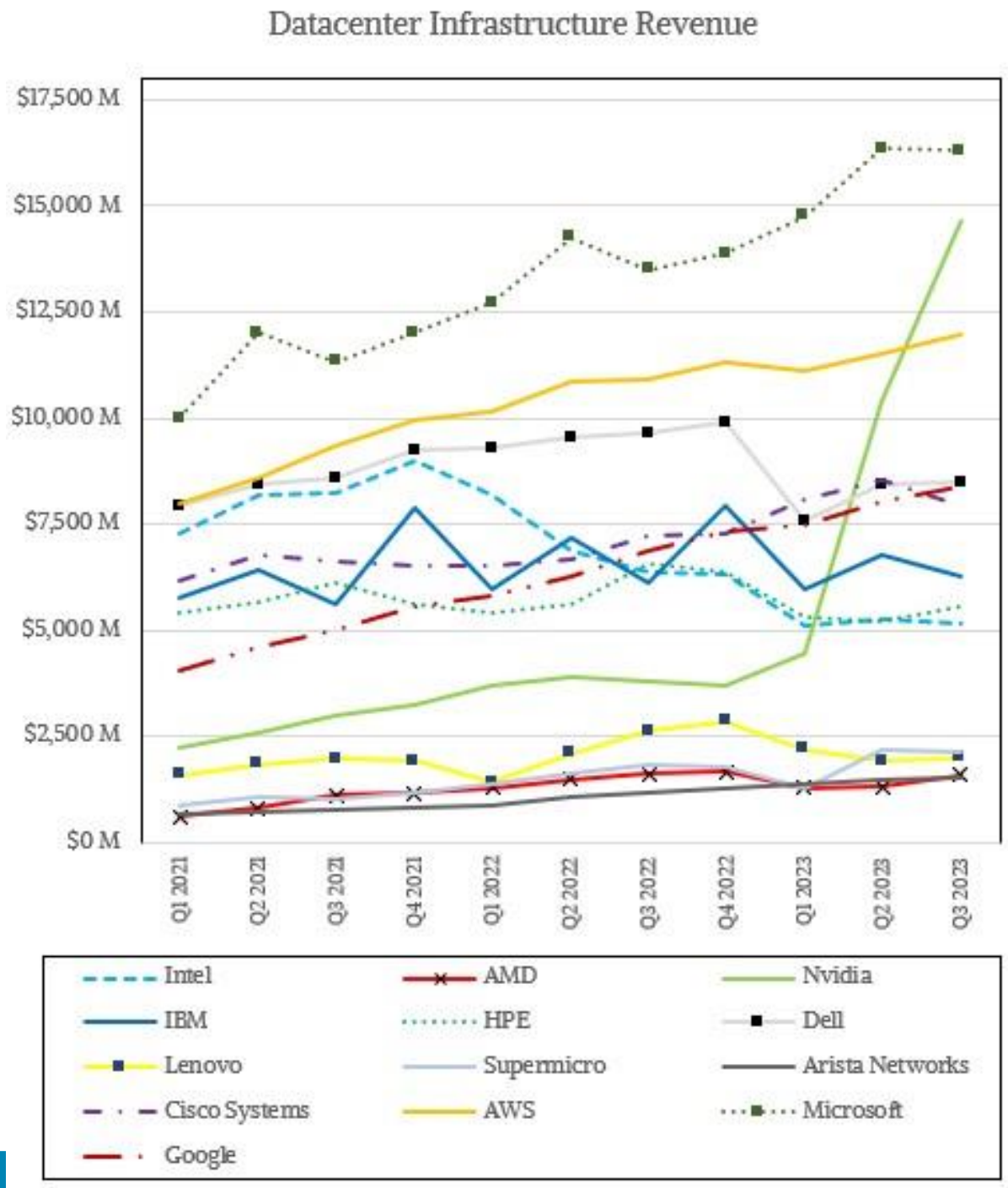
- Clock speed increases have dropped to 1% per year in recent years

- The 22% growth includes the parallelization from multiple cores

Top500 supercomputer performance

GPUs in TOP500 supercomputers




Datacenter Infrastructure Revenue
$17,500 M
$15,000 M
$12,500 M
$10,000 M
$7,500 M
$5,000 M
$2,500 M
$0 M
Q1 2021
Q2 2021
Q3 2021
Q4 2021
Q1 2022
Q2 2022
Q3 2022
Q4 2022
Q1 2023
Q2 2023
Q3 2023
Intel
AMD
Nvidia
IBM
HPE
Dell
Lenovo
Supermicro
Arista Networks
Cisco Systems
AWS
Microsoft
Google

Typical PC (in 1997)

- Clock speed: 50 to 300 MHz.
- Clock cycle time: 3 to 20 ns.
- Cache size: 8 KByte to 512 KByte.
- Cache miss penalty: 5 to 20 cycles.
- CPI (not counting cache miss penalties): 3 to .3
- Main (physical) memory size: 8MByte to 256MByte.
- Virtual address length: 32 or 64 bits.
- Capacity of a disk: 300 MByte to 10 GByte.
- Bandwidth from disk: 1 to 20 MByte/sec.
- Disk latency: 5 to 50 msec.
- Page size: 4 KByte to 64 KByte.
- TLB size: 64 to 1024 entries.

Intel Desktop Technology

End of frequency scaling

End of core scaling

End of technology scaling?

Year	Processor	Transistors	Transistor Size	Performance
1982	i286	~134,000	1.5 mm	**6 - 25 MHz**
1986	i386	~270,000	1 mm	**16 – 40 MHz**
1989	i486	~1 million	.8 mm	**16 - 133 MHz**
1993	Pentium	~3 million	.6 mm	**60 - 300 MHz**
1995	Pentium Pro	~4 million	.5 mm	**150 - 200 MHz**
1997	Pentium II	~5 million	.35 mm	**233 - 450 MHz**
1999	Pentium III	~10 million	.25 mm	**450 – 1400 MHz**
2000	Pentium 4	~50 million	180 nm	**1.3 – 3.8 GHz**
2005	Pentium D	~200 million	90 nm	**2 cores/package**
2006	Core 2	~300 million	65 nm	**2 cores/chip**
2008	"Nehalem"	~800 million	45 nm	**4 cores/chip**
2010-11	"Westmere" / "Sandy Bridge"	~1.2 billion	32 nm	**6 cores/chip**
2012-13	"Ivy Bridge" / "Haswell"	~1.4 billion	22 nm	**8 cores/chip**
2014-15	"Broadwell" / "Skylake"	~2.8 billion	**14 nm**	**8 cores/chip**
2016-18	"Kaby Lake" / "Coffee Lake" / "Whiskey Lake"	~5.6 billion	**14 nm (!!!)**	**8 cores/chip**
2018?	"Cannon Lake"	?	**???**	**?**

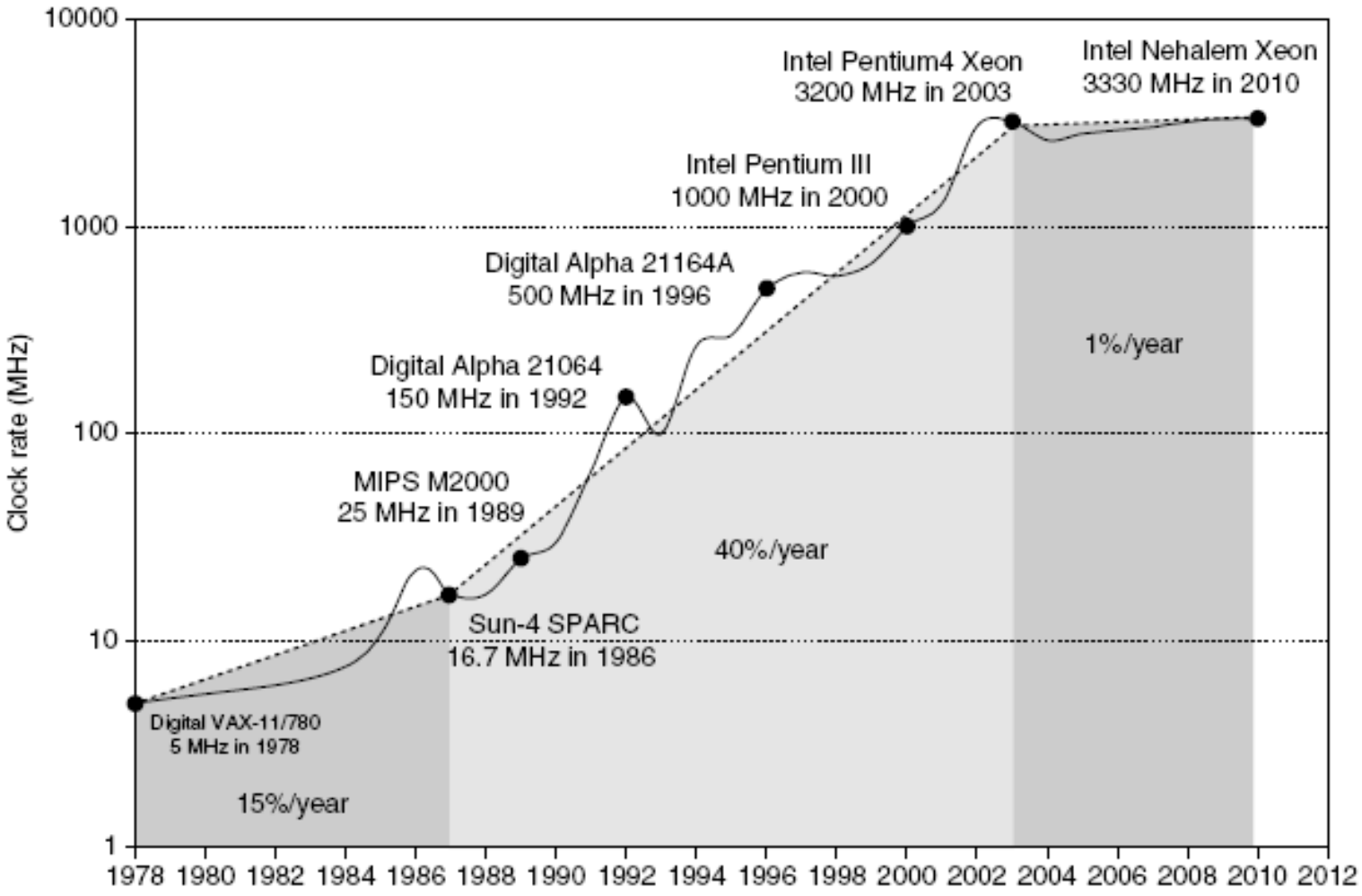
Courtesy: Jason D. Bakos

Performance: Intel Desktop Peak Performance

Desktop Processor Generation	Max. Clock Speed (GHz)	Peak Integer IPC	Max. Number of Cores	Max. DRAM Bandwidth (GB/s)	Peak Floating Point (Gflop/s)	L3 cache (MB)
Core (2006)	3.33	4	4	25.6	107	8
Penryn (2007)	3.33	4	4	25.6	107	8
Westmere (2010)	3.60	4	6	25.6	173	12
Ivy Bridge (2013)	3.70	6	6	25.6	355	15
Broadwell (2015)	3.80	8	6	25.6	365	30
Kaby Lake (2017)	4.00	8	6	42.7	768	32

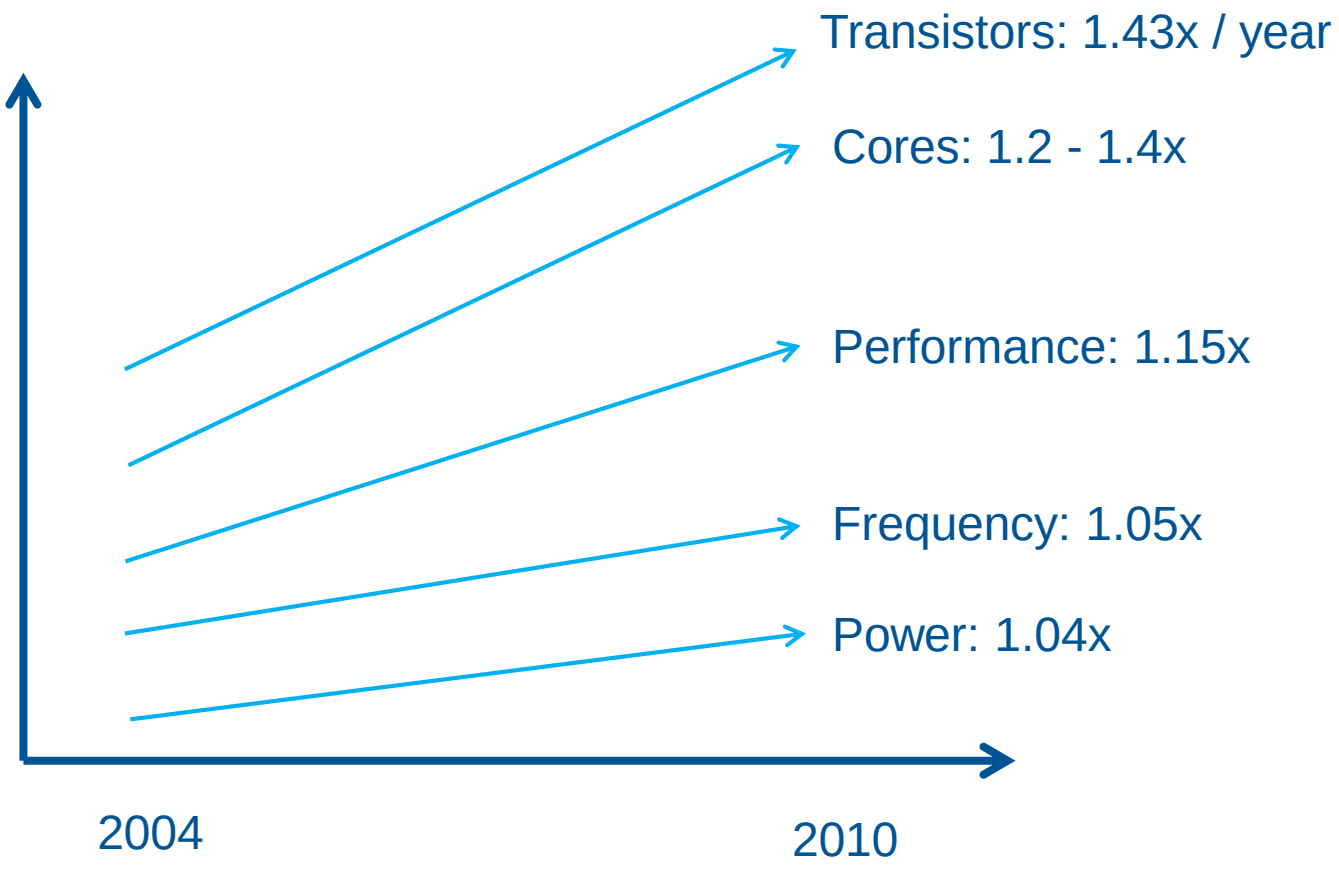
Courtesy: Jason D. Bakos

Clock Speed Increases

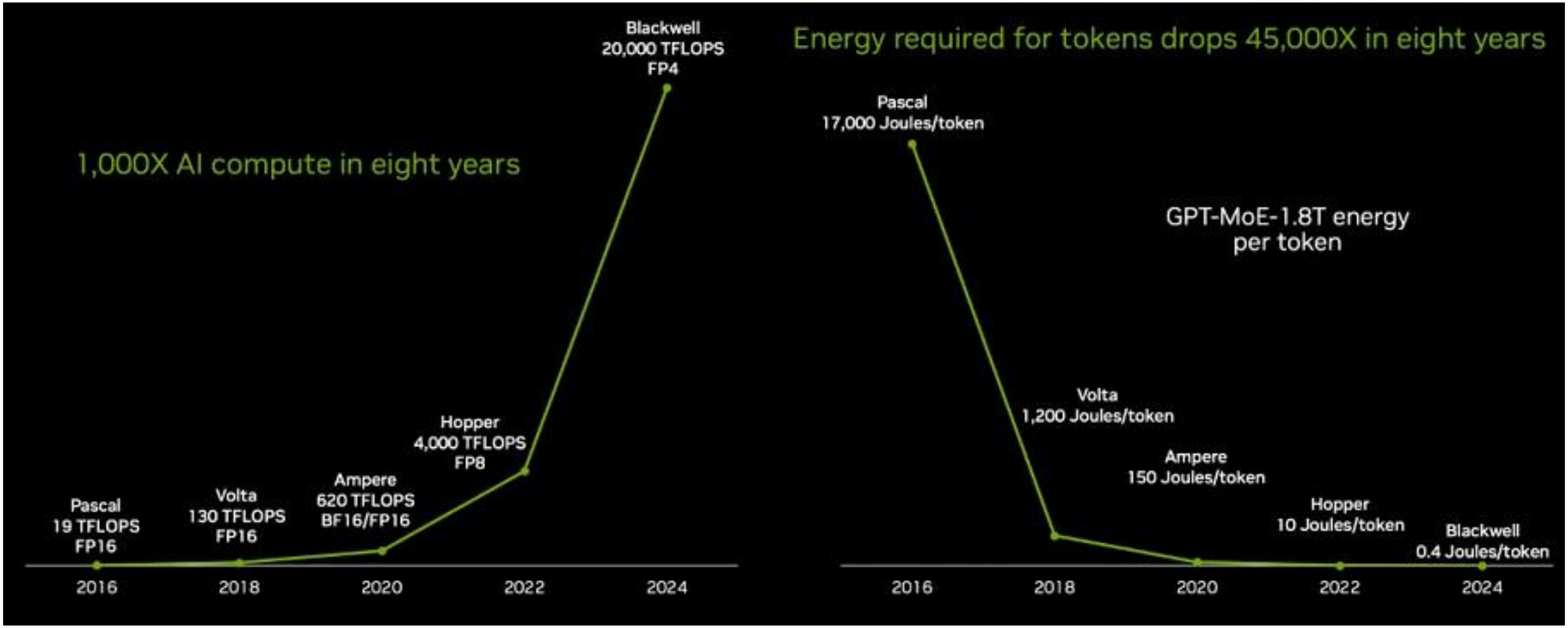


Source: H&P textbook

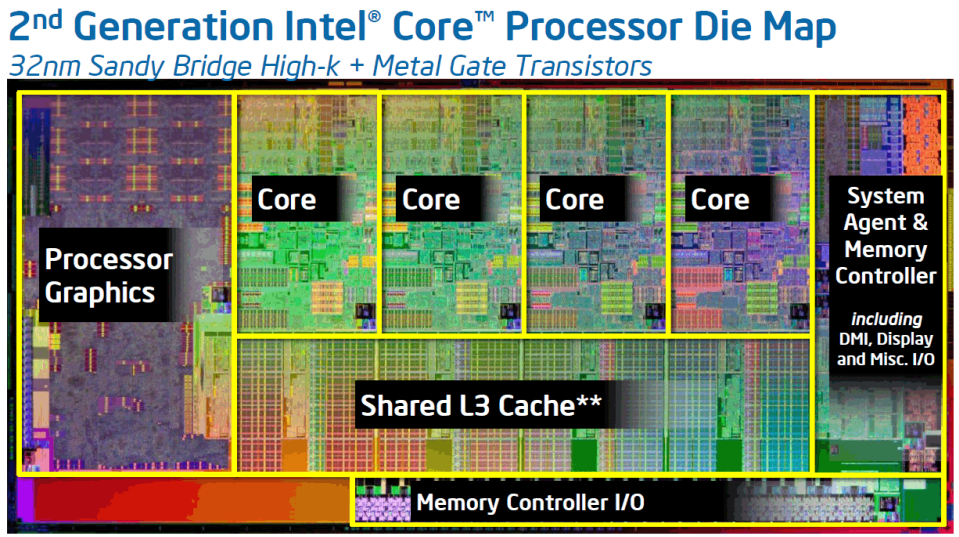
Recent Microprocessor Trends

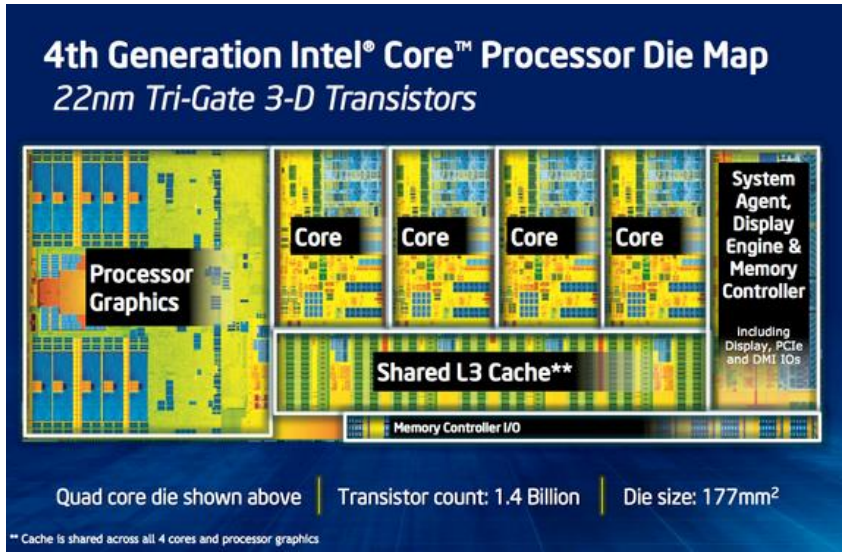


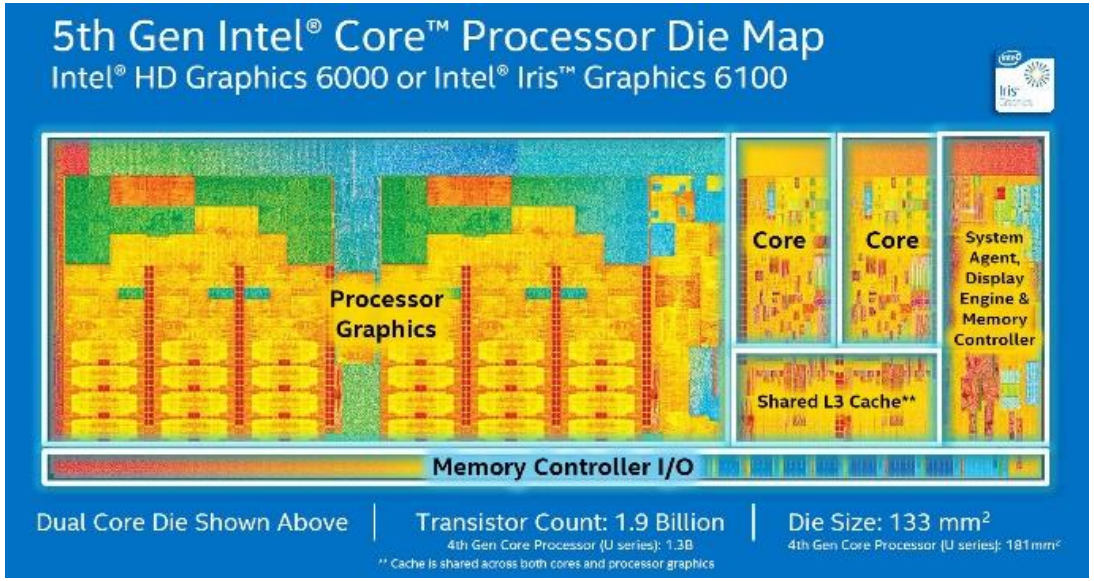
Source: Micron University Symp.


1,000X AI compute in eight years
Pascal
19 TFLOPS
FP16
Volta
130 TFLOPS
FP16
Ampere
620 TFLOPS
BF16/FP16
Hopper
4,000 TFLOPS
FP8
Blackwell
20,000 TFLOPS
FP4
2016
2018
2020
2022
2024
Energy required for tokens drops 45,000X in eight years
GPT-MoE-1.8T energy
per token
Pascal
17,000 Joules/token
Volta
1,200 Joules/token
Ampere
150 Joules/token
Hopper
10 Joules/token
Blackwell
0.4 Joules/token
2016
2018
2020
2022
2024

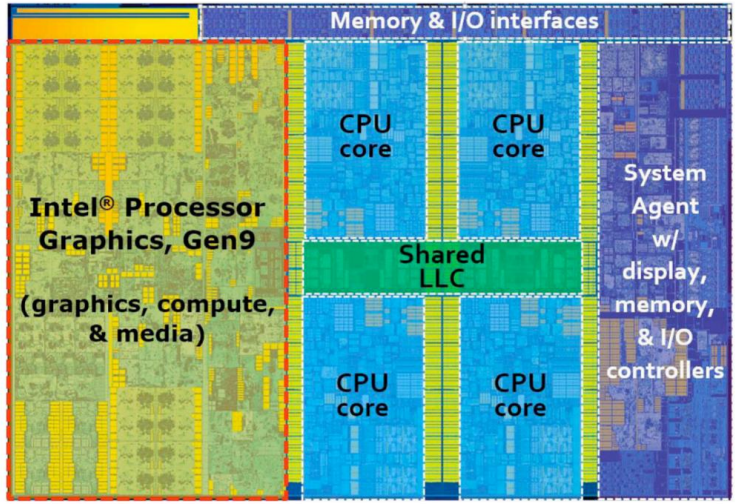
Why Learn Hardware Design?







7th gen Skylake



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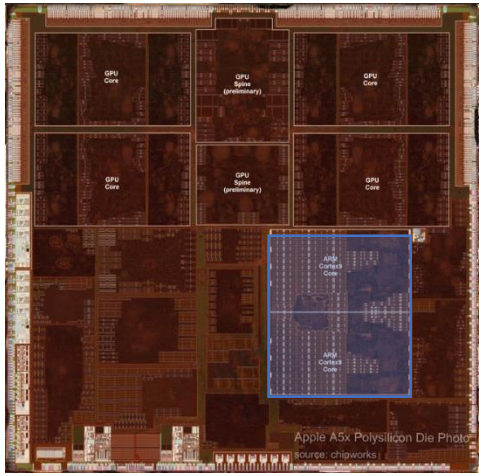
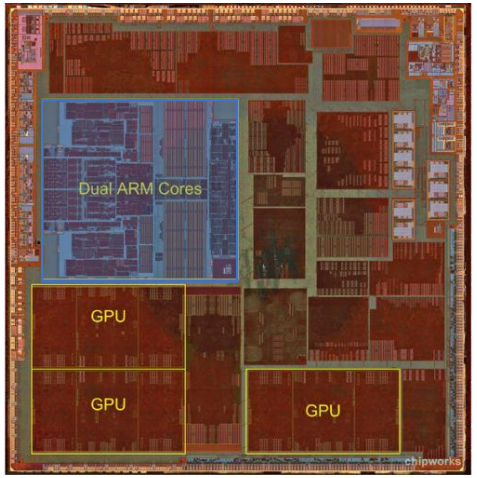
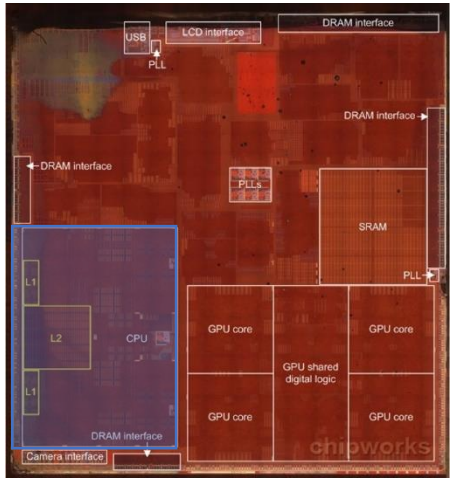


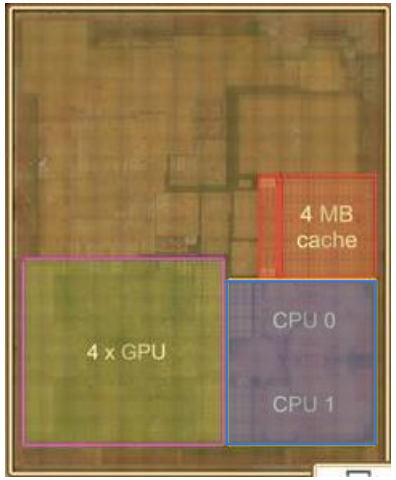
Apple A5x



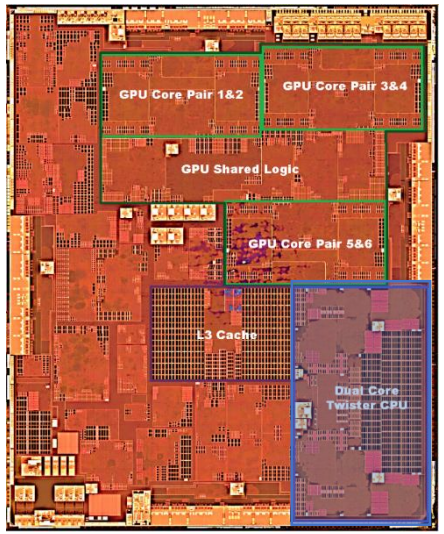
Apple A6



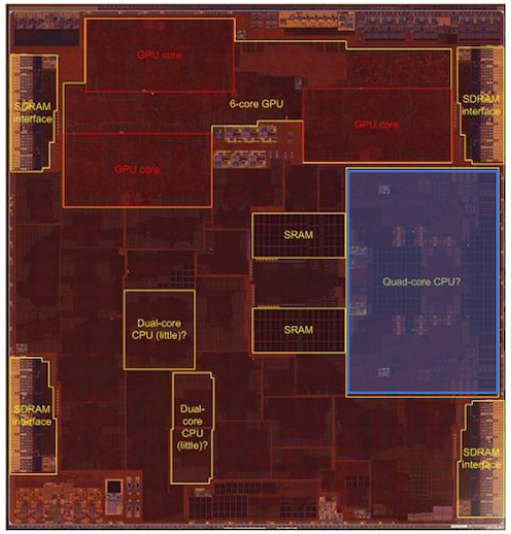
Apple A7



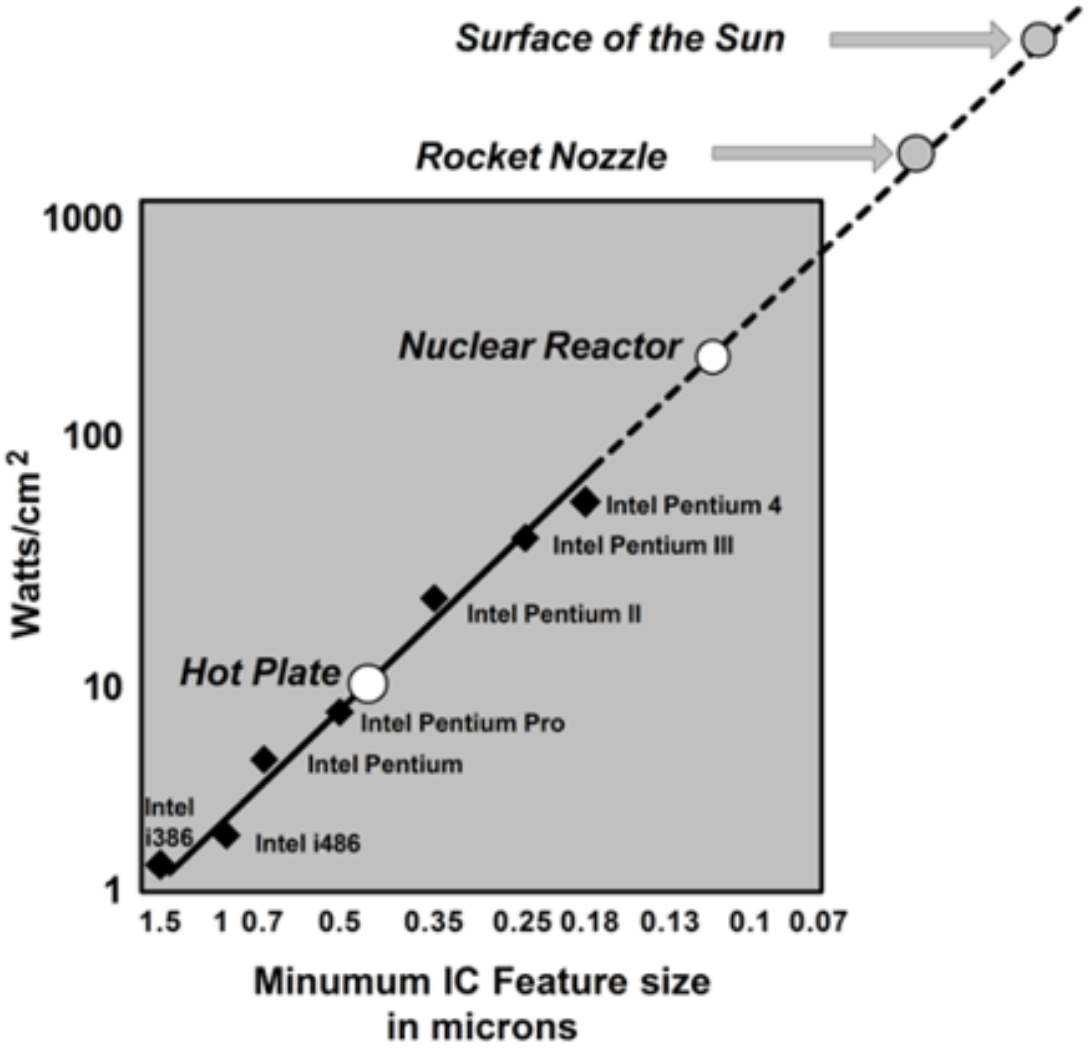
Apple A8



Apple A9



Apple A10


Surface of the Sun
Rocket Nozzle
Nuclear Reactor
Hot Plate
Intel Pentium 4
Intel Pentium III
Intel Pentium II
Intel Pentium Pro
Intel Pentium
Intel i486
Intel i386
1000
100
10
1
Watts/cm 2
1.5 1 0.7 0.5 0.35 0.25 0.18 0.13 0.1 0.07
Minumum IC Feature size in microns

Cost of performance in supercomputers

- Inflation adjusted budgets for the machines at Oak Ridge for the past decade against their peak FP64 performance:
- The 1.75 petaflops “Jaguar” system cost $82 million per petaflops
- The 17.6 petaflops “Titan” system cost $6.5 million per petaflops
- The 200 petaflops “Summit” machine cost a little more than $1 million per petaflops
- The new 2 exaflops “Frontier” machine cost $400,000 per petaflops.